Binary cross-detection (N = 40 training clips per class; detected = p >= 0.5)

BirdNET 2.4

Binary detector (training clips per class)	Device static (50)	Generator (50)	Traffic (50)	Engine (50)	Power tools (29)	Human vocal (50)	Metal clanging (30)	Airplane (25)	Gun (9)	Fireworks (6)	Human non-vocal (3)	Wind (50)	Thunder (8)	Water (5)
Device static	.66	.30	.17	.06	.13	.07	.11	.46	.29	.15	.00	.21	.30	.72
Generator	.03	.32	.05	.09	.03	.07	.05	.13	.00	.00	.03	.13	.21	.30
Traffic	.29	.53	.58	.55	.21	.12	.37	.84	.00	.35	.00	.21	.21	.26
Engine	.22	.16	.32	.90	.28	.07	.22	.66	.03	.05	.30	.14	.16	.00
Power tools	.00	.03	.00	.00	.11	.07	.02	.00	.00	.00	.00	.01	.00	.00
Human vocal	.00	.01	.00	.00	.03	.30	.06	.00	.00	.00	.17	.00	.00	.00
Metal clanging (N=20)	.38	.23	.10	.00	.48	.16	.27	.26	.98	1.00	.33	.54	.28	.34
Airplane (N=20)	.11	.15	.36	.65	.14	.04	.14	.50	.02	.43	.13	.10	.15	.00
Gun (N=10)	.11	.06	.13	.00	.25	.01	.12	.04	.93	.78	.00	.00	.00	.14
Fireworks (N=5)	.08	.07	.15	.02	.32	.13	.18	.08	.84	.87	.03	.10	.05	.08
Human non-vocal (N=5)	.01	.03	.02	.01	.07	.11	.04	.05	.00	.00	.07	.00	.04	.04
Wind	.34	.24	.07	.01	.11	.01	.13	.14	.10	.60	.00	.96	.72	.00
Thunder (N=10)	.09	.07	.07	.00	.03	.00	.14	.09	.00	.08	.00	.49	.86	.00
Water (N=5)	.07	.19	.22	.34	.24	.38	.27	.14	.03	.10	.43	.08	.19	.50

Test clips of class (n)

BirdNET 3.0

Binary detector (training clips per class)	Device static (50)	Generator (50)	Traffic (50)	Engine (50)	Power tools (29)	Human vocal (50)	Metal clanging (30)	Airplane (25)	Gun (9)	Fireworks (6)	Human non-vocal (3)	Wind (50)	Thunder (8)	Water (5)
Device static	.56	.32	.31	.25	.28	.42	.30	.58	.17	.23	.17	.10	.23	.94
Generator	.22	.42	.15	.08	.21	.18	.11	.11	.00	.00	.00	.12	.01	.58
Traffic	.25	.39	.64	.59	.36	.22	.31	.59	.29	.42	.57	.15	.29	.28
Engine	.21	.27	.49	.81	.20	.13	.20	.50	.36	.27	.83	.39	.60	.08
Power tools	.10	.06	.00	.05	.12	.08	.03	.00	.00	.00	.00	.06	.00	.00
Human vocal	.32	.13	.19	.17	.13	.48	.17	.20	.14	.13	.00	.20	.07	.00
Metal clanging (N=20)	.42	.27	.12	.08	.37	.51	.30	.14	.76	1.00	.33	.64	.30	.00
Airplane (N=20)	.29	.21	.40	.62	.19	.12	.23	.40	.41	.13	.00	.22	.01	.00
Gun (N=10)	.37	.46	.51	.33	.50	.35	.50	.65	.57	.70	.57	.01	.41	1.00
Fireworks (N=5)	.09	.44	.53	.31	.48	.33	.52	.41	.14	.38	.50	.17	.68	.78
Human non-vocal (N=5)	.06	.06	.10	.00	.14	.12	.09	.10	.03	.00	.00	.00	.00	.12
Wind	.30	.24	.12	.16	.14	.12	.16	.14	.01	.00	.70	.93	.93	.00
Thunder (N=10)	.24	.17	.20	.45	.04	.09	.13	.23	.07	.12	.30	.60	.68	.06
Water (N=5)	.50	.46	.48	.57	.64	.63	.58	.58	.72	.88	.63	.47	.79	.86

Test clips of class (n)

Perch v8

Binary detector (training clips per class)	Device static (50)	Generator (50)	Traffic (50)	Engine (50)	Power tools (29)	Human vocal (50)	Metal clanging (30)	Airplane (25)	Gun (9)	Fireworks (6)	Human non-vocal (3)	Wind (50)	Thunder (8)	Water (5)
Device static	.68	.43	.24	.12	.12	.11	.27	.45	.16	.12	.00	.38	.36	.60
Generator	.12	.32	.06	.14	.16	.09	.06	.12	.02	.00	.23	.19	.28	.04
Traffic	.51	.44	.54	.41	.17	.08	.24	.66	.02	.28	.20	.26	.41	.08
Engine	.25	.41	.34	.77	.30	.10	.23	.48	.26	.17	.37	.36	.54	.10
Power tools	.15	.12	.00	.03	.13	.19	.06	.00	.00	.00	.00	.16	.00	.00
Human vocal	.00	.09	.05	.01	.09	.69	.10	.01	.00	.00	.00	.00	.10	.06
Metal clanging (N=20)	.41	.26	.21	.23	.52	.14	.45	.21	1.00	1.00	.63	.42	.31	.42
Airplane (N=20)	.39	.21	.56	.58	.19	.15	.22	.51	.33	.15	.00	.12	.04	.02
Gun (N=10)	.10	.32	.34	.15	.54	.14	.53	.24	.78	.73	.00	.05	.05	.90
Fireworks (N=5)	.06	.22	.38	.02	.39	.23	.52	.22	.69	.58	.13	.09	.06	.70
Human non-vocal (N=5)	.45	.13	.21	.20	.15	.14	.16	.32	.01	.03	.43	.04	.00	.02
Wind	.48	.20	.09	.06	.01	.19	.16	.02	.00	.28	.00	.95	.66	.02
Thunder (N=10)	.82	.32	.22	.29	.18	.28	.24	.12	.33	.67	.00	.80	.97	.12
Water (N=5)	.33	.46	.45	.66	.79	.47	.57	.30	.80	.82	1.00	.35	.59	.56

Test clips of class (n)

Perch 2.0

Binary detector (training clips per class)	Device static (50)	Generator (50)	Traffic (50)	Engine (50)	Power tools (29)	Human vocal (50)	Metal clanging (30)	Airplane (25)	Gun (9)	Fireworks (6)	Human non-vocal (3)	Wind (50)	Thunder (8)	Water (5)
Device static	.73	.29	.26	.16	.04	.15	.09	.61	.16	.17	.00	.36	.60	.88
Generator	.10	.35	.10	.13	.08	.10	.10	.18	.00	.00	.00	.15	.05	.24
Traffic	.27	.51	.58	.54	.35	.10	.31	.76	.00	.22	.47	.09	.30	.20
Engine	.24	.37	.37	.82	.31	.18	.28	.57	.01	.00	.33	.04	.09	.16
Power tools	.05	.08	.00	.03	.22	.09	.02	.00	.00	.00	.00	.00	.00	.00
Human vocal	.00	.11	.08	.00	.20	.71	.13	.02	.00	.00	.37	.00	.00	.06
Metal clanging (N=20)	.29	.15	.12	.04	.56	.22	.57	.03	1.00	1.00	.20	.54	.19	.04
Airplane (N=20)	.36	.27	.52	.47	.17	.22	.22	.44	.00	.00	.00	.03	.00	.00
Gun (N=10)	.06	.27	.24	.03	.48	.15	.57	.17	1.00	1.00	.00	.01	.00	.60
Fireworks (N=5)	.01	.11	.24	.01	.43	.12	.44	.18	.97	.95	.03	.02	.04	.20
Human non-vocal (N=5)	.00	.09	.04	.00	.17	.03	.22	.04	.00	.00	.57	.00	.00	.00
Wind	.30	.20	.08	.06	.11	.12	.18	.21	.04	.05	.17	.94	.88	.28
Thunder (N=10)	.37	.16	.15	.09	.26	.16	.34	.20	.57	.85	.03	.97	1.00	.24
Water (N=5)	.07	.23	.23	.34	.57	.23	.48	.34	.16	.38	.93	.05	.20	.46

Test clips of class (n)

CNN14 (AudioSet)

Binary detector (training clips per class)	Device static (50)	Generator (50)	Traffic (50)	Engine (50)	Power tools (29)	Human vocal (50)	Metal clanging (30)	Airplane (25)	Gun (9)	Fireworks (6)	Human non-vocal (3)	Wind (50)	Thunder (8)	Water (5)
Device static	.53	.22	.19	.15	.08	.05	.14	.36	.04	.15	.00	.07	.06	.54
Generator	.13	.22	.09	.11	.15	.03	.10	.04	.00	.00	.00	.03	.00	.06
Traffic	.42	.52	.65	.59	.34	.16	.18	.80	.00	.07	.00	.17	.44	.18
Engine	.38	.26	.38	.92	.35	.01	.23	.67	.00	.22	.23	.31	.65	.00
Power tools	.05	.03	.00	.03	.13	.06	.00	.00	.00	.00	.00	.00	.00	.00
Human vocal	.00	.07	.04	.00	.11	.67	.01	.00	.00	.00	.23	.00	.00	.00
Metal clanging (N=20)	.36	.10	.12	.07	.50	.39	.56	.04	.39	.90	.53	.32	.06	.00
Airplane (N=20)	.53	.41	.61	.84	.41	.15	.20	.91	.00	.20	.07	.02	.05	.00
Gun (N=10)	.08	.03	.06	.00	.13	.05	.22	.00	1.00	.90	.00	.02	.00	.00
Fireworks (N=5)	.02	.00	.01	.00	.00	.01	.08	.00	.69	.13	.00	.00	.00	.00
Human non-vocal (N=5)	.13	.01	.06	.05	.16	.34	.08	.03	.00	.00	.67	.00	.00	.00
Wind	.18	.10	.05	.05	.00	.05	.07	.00	.00	.20	.00	.98	.99	.02
Thunder (N=10)	.18	.10	.07	.15	.05	.04	.19	.02	.00	.58	.03	1.00	.96	.00
Water (N=5)	.06	.10	.03	.05	.17	.11	.20	.00	.00	.00	.07	.01	.00	.18

Test clips of class (n)

ResNet-18 (ImageNet)

Binary detector (training clips per class)	Device static (50)	Generator (50)	Traffic (50)	Engine (50)	Power tools (29)	Human vocal (50)	Metal clanging (30)	Airplane (25)	Gun (9)	Fireworks (6)	Human non-vocal (3)	Wind (50)	Thunder (8)	Water (5)
Device static	.64	.29	.16	.13	.10	.11	.13	.31	.10	.15	.00	.30	.15	.24
Generator	.04	.26	.18	.36	.07	.03	.02	.04	.02	.00	.00	.04	.00	.00
Traffic	.44	.29	.55	.65	.23	.17	.09	.88	.24	.40	.00	.14	.56	.00
Engine	.46	.32	.44	.89	.39	.15	.25	.93	.77	.37	.33	.14	.50	.00
Power tools	.06	.24	.03	.00	.31	.28	.21	.00	.00	.00	.70	.00	.00	.26
Human vocal	.20	.09	.11	.15	.24	.55	.33	.13	.01	.05	.10	.19	.41	.04
Metal clanging (N=20)	.42	.23	.10	.03	.57	.44	.53	.19	.40	.82	.27	.89	.68	.18
Airplane (N=20)	.30	.27	.47	.84	.16	.15	.14	.87	.38	.37	.10	.04	.35	.00
Gun (N=10)	.21	.07	.17	.24	.21	.08	.16	.48	.59	.95	.00	.05	.40	.00
Fireworks (N=5)	.24	.22	.36	.17	.27	.44	.42	.26	.46	.87	.03	.37	.42	.12
Human non-vocal (N=5)	.04	.09	.04	.12	.38	.26	.27	.03	.01	.00	1.00	.02	.01	.08
Wind	.37	.22	.06	.00	.01	.14	.16	.15	.02	.70	.00	.96	.84	.02
Thunder (N=10)	.22	.16	.09	.11	.05	.01	.14	.22	.57	.83	.00	.85	1.00	.00
Water (N=5)	.20	.35	.32	.31	.66	.51	.49	.20	.09	.10	1.00	.24	.09	.44

Test clips of class (n)

Proportion detected; diagonal = recall, mean of 10 draws

1.0
0.8
0.6
0.4
0.2
0.0